Time Definition in Ontology:

The Sequence of Snapshots of Spatial Energy Distribution

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The core thesis of this paper is grounded in the framework of Spatial Geometric Realism proposed by the author in 2025 [13]. Building upon that earlier work, the present paper reorganizes its arguments from a metaphysical perspective, with the aim of engaging in dialogue with the central concerns of contemporary philosophy of time.The thermodynamic arrow of time, its irreversibility, and the method for computing the thermodynamic entropy of each snapshot are addressed in detail in the main text[13].

Abstract

The concept of time has always occupied a central position in both philosophy and physics, yet its definition has long suffered from a logical circularity—defining time by clocks and clocks by time. This paper attempts to reconstruct a definition of time from first principles that avoids this circularity. The basic premises are: time is the measure of change; the object of change is the universe as a whole; the universe is defined as the closed system of all that exists; its fundamental ontology is a vacuum with fixed topological adjacency relations; any transformation of the universe can be mapped as a redistribution of energy over a fixed topological grid; global objective time is the discrete sequence of panoramic snapshots of this energy distribution; and subjective time is the perception of this sequence via finite sampling rates. This framework is compared with the time concepts of Newton, Leibniz, Mach, Bridgman's operationalism, and Whitehead's process philosophy. Recent discussions on vacuum topology, emergent time, and quantum thermodynamics are cited to demonstrate its advantages in logical consistency, ontological clarity, and compatibility with established physical theories.

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1. Introduction: The Definitional Dilemma of Time

What is time? This question has pervaded the history of Western philosophy and physics. Newton, in his Philosophiæ Naturalis Principia Mathematica, posited absolute time, holding that it "flows equably without relation to anything external". Leibniz, in contrast, maintained that time is nothing more than the order of events, possessing no independent reality apart from the sequence of occurrences . This debate was profoundly informed by Einstein's special and general theories of relativity, yet it has not been settled at the philosophical level.

In the 20th century, the process philosophy tradition, most notably that of Whitehead, further emphasised that the nature of reality is not static substance but the becoming of "actual occasions"; time is not a container but the very rhythm of this becoming. This perspective resonates subtly with the "problem of time" in contemporary quantum gravity—if the fundamental substrate is timeless, whence does time arise?

This paper does not aim to rehearse the historical contours of these debates. Instead, it confronts a straightforward logical question: if time is the measure of change, then exactly what change does it measure? This deceptively simple question has long remained unanswered. Physics typically defines time by "clock readings," but clocks themselves require the concept of time for their own definition, thus generating a circularity. Operationalists attempt to circumvent this by asserting that "time is what clocks measure", but this offers no independent definition, merely substituting one instrument for the definiendum.

This paper attempts to reconstruct, from ontological first principles, a definition of time that avoids circularity and remains compatible with established physics.

2. Operationalism and Its Logical Limitations

Operationalism, as systematically articulated by Bridgman, holds that the meaning of a concept is equivalent to the set of operations used to measure it. In the context of time, this implies that "time" should be defined as "that which is measured by clocks." This stance has become dominant in physics, since the aim of physics is to predict experimental outcomes, not to pursue metaphysical essences.

However, operationalism faces logical difficulties here:

(1) Circularity: A clock is defined as "an instrument for measuring time," while time is defined as "what clocks measure." Each term depends on the other, providing no independent foundation.

(2) Pars pro toto: The universe contains far more than clock changes. Stellar nucleosynthesis, galactic evolution, biological aging—these are not clock readings, yet they are typically considered to occur "in time." If time were merely clock change, then when clocks stop, would cosmic change also cease? That is evidently absurd.

(3) Avoidance of ontology: Operationalism sidesteps the fundamental question

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"What is the object of change?" by replacing it with instrumental proxies.

The stance adopted here is that operationalism is a valid measurement strategy but not a satisfactory philosophical definition. To understand time at the ontological level, one must directly answer: "What change does time measure?"

3. Ontological Presuppositions: The Closed Universe and Vacuum as Fundamental Substance

To define time, one must first specify the object of change. This paper adopts the following ontological framework:

3.1 The Universe as a Closed System

Define the universe as "the totality of all that exists." If anything were described as "outside the universe," it would still fall within the scope of "all that exists," and hence remain part of the universe. It follows that the universe is necessarily closed; there exists no external reference frame. This definition is logically akin to Spinoza's concept of substance—that which is self-caused and independent of anything else.

3.2 Vacuum as the Sole Fundamental Ontology

From the mass-energy equivalence $E = m c^2$, matter can be transformed into energy, and energy can be dissipated into the vacuum (as in quantum field theoretic vacuum fluctuations and dark energy). Thus, matter is not a fundamental entity but a condensed form of energy. The vacuum—understood as the ground state of quantum fields—manifests physical effects (Casimir effect, Lamb shift, etc.) and is ontologically more basic.

It follows that the vacuum is the sole fundamental ontological reality; matter, dark matter, and radiation are all excitations of the vacuum. This monistic position is consistent with the ontological interpretation of quantum field theory and with contemporary cosmology in which dark energy is identified with vacuum energy density. Recent discussions on vacuum topology and the origin of matter have suggested that elementary particles may be viewed as topological defects of the vacuum manifold, a direction closely aligned with the "matter = vacuum excitation" hypothesis adopted here.

3.3 Fixed Topological Adjacency Relations

Observations have revealed no singularities, wormholes, traversable time-like closed curves, or other exotic topological structures. The smooth curvature of light around black holes indicates that the local adjacency relations of the spacetime manifold remain continuous and unbroken at the classical level. This paper therefore posits as an axiom: the topological adjacency relations of the cosmic vacuum are fixed and invariant. It excludes the possibility of time travel and singularities, thereby simplifying the treatment of the arrow of time.

4. A Dual Definition of Time

Based on the ontological framework above, this paper proposes a dual definition of time.

4.1 Global Objective Time

The state of the universe at any given instant can be represented as the distribution

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of energy density/field values at each point on a spatial topological grid—that is, a "panoramic snapshot of cosmic energy distribution." The transformation of the universe consists in the redistribution of energy over this fixed topological adjacency grid—not in matter "moving through" space, since such motion would imply a change in topological adjacency, which is prohibited by our axiom.

Thus we define:

Global objective time = the discrete sequence of panoramic snapshots of cosmic energy distribution.

This sequence is absolute, because it corresponds to no observer's sampling; it is the logical serial number of the evolution of the total cosmic state. The object of change is explicitly "the total cosmic energy distribution," and the content of change is "the configurational difference between snapshots ($\Delta E(x)$)."

This definition avoids Newton's metaphysical assumption of "uniform flow" and also avoids the circularity of operationalism. It can be understood as a globalised version of Leibnizian relational time: whereas Leibniz focused on local event-order relations, this paper extends the ordering relation to the totality of cosmic states.

The "discrete sequence" here should be understood as logical continuity, not temporal metric continuity. Whitehead, in Process and Reality, argued that ultimate reality consists of "prehensions," each of which integrates and responds to preceding states. The "snapshot sequence" in this paper can be viewed as a physicalised version of Whitehead's "actual occasion" sequence—each snapshot prehends all gradient information from its predecessors, thereby driving the generation of the next snapshot. The difference lies in the driver: Whitehead attributed this generation to the lure of "eternal objects," whereas this paper identifies the physical driver as energy gradients.

4.2 Subjective Time

No physical observer (human senses or instruments) can directly perceive the global snapshot sequence; they can only access local information via finite sampling rates. Hence:

Subjective time = the mapping of the local space snapshot sequence by an observer or instrument, based on a finite sampling rate.

The upper limit of the sampling rate is given by the Planck frequency, which is the minimum temporal resolution describable by current physical theory. Subjective time is not global or local objective time itself, but an approximation of it.

Milburn (2020)demonstrates that clocks, as open non-equilibrium systems, are subject to thermodynamic dissipation, implying that they cannot serve as an objective, external backdrop for measuring rates of change—a conclusion that aligns with the present framework's rejection of operationalist definitions of time.

5. Dynamics and the Origin of Quantum Probability

5.1 Gradient-Driven Dynamics

The gradient-driven dynamics proposed here aligns with the thermodynamic formulation of time by Tuisku, Pernu, and Annila (2009), who describe physical evolution as a process driven by energy dispersal.

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Energy redistribution does not occur without cause. This paper posits a dynamical axiom: the driving force behind energy redistribution is the energy gradient between adjacent lattice sites. Energy tends to flow from regions of higher density to regions of lower density—this is the common underlying logic of the diffusion equation, the heat equation, and the Einstein field equations in general relativity, where the energy-momentum tensor on the right-hand side drives curvature change.

The Hamiltonian is thus re-positioned within this framework: the global Hamiltonian is the current panoramic snapshot of energy distribution, containing all local gradient information; the local Hamiltonian is the local snapshot of an approximately closed subsystem.

This gradient-driven model invites an interesting comparison with Whitehead's notion of "creative advance": Whitehead held that each actual occasion is guided by the "objective immortality" of its predecessors; this paper identifies energy gradients as the physical carrier of that guidance, translating a metaphysical description into a measurable variable.

5.2 Ontological Source of Quantum Probability

Since the gradient differences at each local site admit multiple combinatorial configurations, given a current snapshot of energy distribution, there may be multiple possible next snapshots. This implies:

Quantum probability does not arise from incomplete observer information, but from the inherent multiplicity of possibilities in fundamental reality.

This multiplicity is the fundamental rule; deterministic classical trajectories are macroscopic approximations (emergent phenomena after decoherence). This interpretation bears an affinity with certain versions of the many-worlds interpretation, but anchors the origin of probability in "the non-uniqueness of gradient combinations on a topological grid" rather than postulating branching universes.

Consequently, the Schrödinger equation

is reinterpreted as follows: the Hamiltonian operator reads the gradient information of the current snapshot and generates the probability-amplitude distribution of all possible next configurations; time evolution is then the serialisation of this generative process.

6. Conclusion

Starting from the intuitive premise that "time is the measure of change," this paper has constructed a definition of time that avoids circularity through a chain of five axioms:

The object of change is the universe as a whole;

The universe is closed, with no external reference;

The fundamental ontology is a vacuum with fixed topology, whose excitations constitute matter;

Cosmic transformation = redistribution of energy over a fixed topological grid;

Global objective time = the sequence of panoramic snapshots of this energy distribution;

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Subjective time = the local mapping of this sequence via finite sampling.

This framework does not rely on clocks for definition, does not conflict with established physical theories (quantum field theory, general relativity, and thermodynamics are all accommodated), and traces the ontological origin of quantum probability to the non-uniqueness of local gradient combinations.

It is neither a revival of Newtonian absolute time nor a mere repetition of Leibnizian relational time. Rather, it establishes a middle path between operationalist measurement-effectiveness and metaphysical definitional completeness—replacing "uniform flow" or "clock readings" with "cosmic energy-distribution snapshot sequence" as the primary meaning of time.

In dialogue with Whitehead's process philosophy, this paper offers a path toward "physicalising" process thought: actual occasions correspond to snapshots, the process of becoming corresponds to gradient-driven dynamics, and subjective time corresponds to finite prehension as sampling. In contrast with emergent-time theories, the global/subjective dual-layer structure avoids the extreme position of rendering time itself wholly emergent, preserving an ontological bridge between physics' "time parameter" and philosophy's "experience of time."

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